

interpolation\linear.py

```
1 import pandas as pd
2
3
4 def read_excel(file):
5     return pd.read_excel(file)
6
7
8 def find_interval(values, point):
9     values = sorted(values.unique())
10
11     if point < values[0] or point > values[-1]:
12         print(f"\nError: {point} is outside the available range "
13               f"({values[0]} to {values[-1]}).")
14         return None, None
15
16     for i in range(len(values)-1):
17         if values[i] <= point <= values[i+1]:
18             return values[i], values[i+1]
19
20
21 def get_corner_value(df, x, y, z, output):
22     row = df[
23         (df["X1"] == x) &
24         (df["X2"] == y) &
25         (df["X3"] == z)
26     ]
27
28     return float(row.iloc[0][output])
29
30
31 def linear_interpolation(a, b, t):
32     return a * (1 - t) + b * t
33
34
35 def trilinear_interpolation(df, x, y, z, output):
36
37     x0, x1 = find_interval(df["X1"], x)
38     y0, y1 = find_interval(df["X2"], y)
39     z0, z1 = find_interval(df["X3"], z)
40
41     if None in (x0, x1, y0, y1, z0, z1):
42         return None
43
44
45     u = (x - x0) / (x1 - x0)
46     v = (y - y0) / (y1 - y0)
47     w = (z - z0) / (z1 - z0)
48
49     f000 = get_corner_value(df, x0, y0, z0, output)
50     f100 = get_corner_value(df, x1, y0, z0, output)
51     f010 = get_corner_value(df, x0, y1, z0, output)
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52     f110 = get_corner_value(df, x1, y1, z0, output)
53     f001 = get_corner_value(df, x0, y0, z1, output)
54     f101 = get_corner_value(df, x1, y0, z1, output)
55     f011 = get_corner_value(df, x0, y1, z1, output)
56     f111 = get_corner_value(df, x1, y1, z1, output)
57
58     c00 = linear_interpolation(f000, f100, u)
59     c10 = linear_interpolation(f010, f110, u)
60     c01 = linear_interpolation(f001, f101, u)
61     c11 = linear_interpolation(f011, f111, u)
62
63     c0 = linear_interpolation(c00, c10, v)
64     c1 = linear_interpolation(c01, c11, v)
65
66     return linear_interpolation(c0, c1, w)
67
68
69 df = read_excel("data.xlsx")
70
71 print("Enter the input values")
72
73 x = float(input("X1 : "))
74 y = float(input("X2 : "))
75 z = float(input("X3 : "))
76
77 predicted_y1 = trilinear_interpolation(df, x, y, z, "Y1")
78 predicted_y2 = trilinear_interpolation(df, x, y, z, "Y2")
79
80 print("\nPrediction")
81 print("-----")
82 print(f"Y1 = {predicted_y1:.4f}")
83 print(f"Y2 = {predicted_y2:.4f}")
84 print("-----")
```